

The Efficacy of Discrepancy in Computer Graphics

Mayur Patel
Toronto, Ontario, Canada
patelm@acm.org

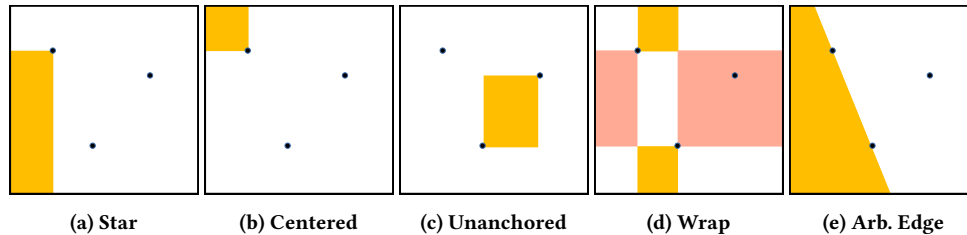


Figure 1: We show regions considered by a selection of discrepancy metrics in two dimensions. In this presentation, we compare discrepancy values of various point sets against the errors found by integrating a variety of functions with those point sets. We report on the best estimators of point set quality in two dimensions.

ABSTRACT

In this presentation, we survey modern discrepancy metrics and use them to estimate the quality of point sets produced from a variety of sample generators in two dimensions. Then, we calculate the actual performance of these point sets for integrating a number of signals in the unit square. Finally, we correlate the estimated performance to the observed result to determine which metrics have the greatest utility as predictors of success for computer graphics applications.

CCS CONCEPTS

• **Computing methodologies** → **Number theory algorithms; Rendering; • Theory of computation** → **Sketching and sampling; Randomness, geometry and discrete structures.**

KEYWORDS

sampling, monte carlo integration, computer graphics, rendering

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1 INTRODUCTION

More than thirty years ago, Shirley conducted a series of experiments to demonstrate how discrepancy metrics can be used to estimate the quality of two-dimensional point sets used for integration during rendering [Shirley 1991]. Since that time, the Star Discrepancy he used has been the standard in computer graphics

for estimating the quality of point set generators even as new discrepancy metrics have emerged. Today, development of new point set generators frequently requires rigorous empirical testing because the Star Discrepancy is not as strong a predictor of real-world performance as one might hope.

In the conclusions of his paper, Shirley recommended, as future work, further investigations into determining which forms of discrepancy would be the best estimators for the quality of point sets. Here, we revisit that question. We survey modern metrics of uniformity and execute empirical tests to determine their efficacy for predicting the results of sampling the unit square with various point sets, a common operation in computer graphics. The findings can be used in the development of new sample generators, to better predict their real-world performance for such applications.

2 METHODOLOGY

We begin by creating sets of sample points. We select a variety of "good" sets and some "bad" ones; and, in some cases, we affect the quality of "good" sets by displacing their samples using various methods. We list the sets of samples we use below:

- ART: Tile-based, blue noise sampler [Ahmed et al. 2017].
- Uniformly Random
- Halton 2,3 Sequence
- Halton 7,11 Sequence
- Halton 21,23 Sequence
- Sobol Sequence
- Sobol Sequence with Owen Scrambling
- Sobol+Owen, displaced with tile noise
- Sobol+Owen, displaced with Voronoi noise
- Sobol+Owen, displaced with value noise
- Sobol+Owen, displaced with simulation noise
- PMJ02: A good stratified sampler [Christensen et al. 2018].

Next, we assemble a variety of functions. The functions are depicted in Figure 2.

We integrate each function for each set of samples using a variety of sample counts. The RMS error across 2048 trials, for each function, is recorded for later analysis.

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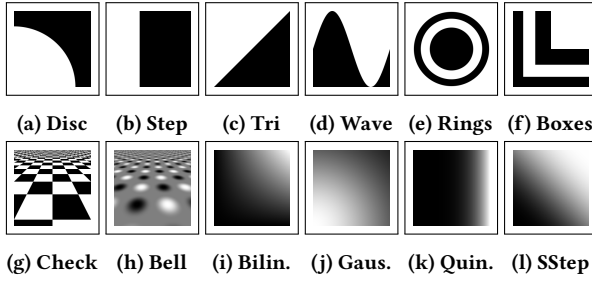


Figure 2: These are functions used for integration testing. While there is some subjectivity in the selection of these signals, it should be noted that subsets of this collection have been used before to test the quality of sample generators (e.g. [Burley 2020; Christensen et al. 2018; Patel 2022; Shirley 1991]).

The counts used are given by S , defined by:

$$S = \{s_i \in \mathbb{Z}^+ : s_i = s_{i-1} + \max(1, \lfloor s_{i-1}/16 \rfloor), s_0 = 4, s_i < 8192\} \quad (1)$$

It is useful to skip some sample counts, particularly with large sample sets, in order to create greater contrast in the correlation calculations we execute later. The sample sizes we test result in a total of 124 tests for each sample set against each function.

Also, we identify uniformity metrics that we wish to test. A full description of each one is beyond the scope of this short presentation, but we provide references for further study.

- Star Discrepancy [Doerr et al. 2014]
- L_2 Discrepancy [Doerr et al. 2014]
- Unanchored L_2 Discrepancy [Morokoff and Caffisch 1994]
- Generalized L_2 Discrepancy [Hickernell 1998a]
- Symmetric L_2 Discrepancy [Hickernell 1998a]
- Centered L_2 Discrepancy [Hickernell 1998a]
- Arbitrary Edge Discrepancy [Dobkin and Mitchell 1993]
- Mixture Discrepancy [Zhou et al. 2013]
- Wrap Discrepancy [Hickernell 1998b]
- Diaphony [Hellekalek and Niederreiter 1998]

To compare the performance of each uniformity metric against the observed integration error, we use Spearman’s Correlation. As a reminder, Spearman’s Correlation uses the rank of each value relative to the full set, as opposed to its actual value, in its calculation. As a result, it can be applied across uniformity metrics to judge the quality of the results in a relative manner. To be clear, all the integrations, across all sample sets and number of samples, are considered together for each function. This provides a reasonable mass of data from which to draw our conclusions. We derive a quality report for each uniformity metric relative to each function.

3 RESULTS AND CONCLUSIONS

We observed how each uniformity metric performed, ranking each metric for each function. We summarize the results of these rankings in the table below.

Metric	Median Rank	Ave Rank	Min Rank	Max Rank	Std Dev
Star Discr.	4	3.83	3	5	0.72
L_2 Discr.	8	8.42	8	9	0.51
Unanchored L_2 Discr.	2	2.42	2	4	0.67
Generalized L_2 Discr.	9	9.08	8	10	0.90
Symmetric L_2	6	5.67	5	6	0.49
Centered L_2	7	7.67	6	9	1.07
Arb. Edge	10	7.92	3	10	2.81
Mixture Discr.	4	4.25	2	7	2.05
Wrap Discr.	1	2.42	1	5	1.83
Diaphony	3	3.33	1	7	2.39

While the Wrap Discrepancy was frequently the best, when it wasn’t, its performance was mediocre. This can be seen by its high standard deviation. On the other hand, the Unanchored L_2 Discrepancy was a more consistent high-performer. It was usually the second-best estimator, and its rank was consistent as seen by its low standard deviation. The Star Discrepancy wasn’t a particularly remarkable estimator, as it was never better than third-best for any of the signals.

Certainly, there is a degree of subjectivity in the selection of sample sets and functions we used for testing. Nonetheless, because the Unanchored L_2 Discrepancy seemed to perform consistently across all our tests, we recommend it as a good general-purpose estimator for point set quality for computer graphics applications.

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